

Retail Sales Performance Tracking and Analysis

TCS RIO— Industry Project

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- **Project Objective**

- To analyze retail sales data and evaluate overall business performance using key performance indicators.
- To identify sales trends over time and understand seasonal variations in customer purchasing behavior.
- To assess product category performance and identify revenue-driving segments.
- To analyze customer demographics to support data-driven business and marketing decisions.

Dataset Overview

- The dataset consists of **1,000 retail transaction records** collected over a defined time period.
- It includes information such as order date, product category, quantity sold, and sales amount.
- Customer attributes like age and gender are included for demographic analysis.
- The data was cleaned to remove inconsistencies and ensure accurate analysis.
- Additional derived fields such as month and age group were created to support trend and segmentation analysis.

KPI Summary

Key Performance Indicators (KPIs)

- **Total Sales (₹4,56,000):**
Represents the total revenue generated from all retail transactions and reflects overall business performance.
- **Average Transaction Value (₹456):**
Measures the average revenue per transaction, providing insights into customer spending patterns and pricing effectiveness.
- **Total Quantity Sold (2,514 units):**
Indicates the total number of products sold across all categories and helps assess sales volume and demand.
- **KPI Importance:**
These metrics provide a quick snapshot of business health and support informed decision-making.

Sales Trend Analysis

- Monthly sales trends were analyzed to understand changes in revenue over time.
- Sales show fluctuations, with peak performance observed during specific periods.
- Declines in certain months indicate the impact of seasonality and customer buying cycles.
- Trend analysis helps in sales forecasting and planning future business strategies.

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Product Category Analysis

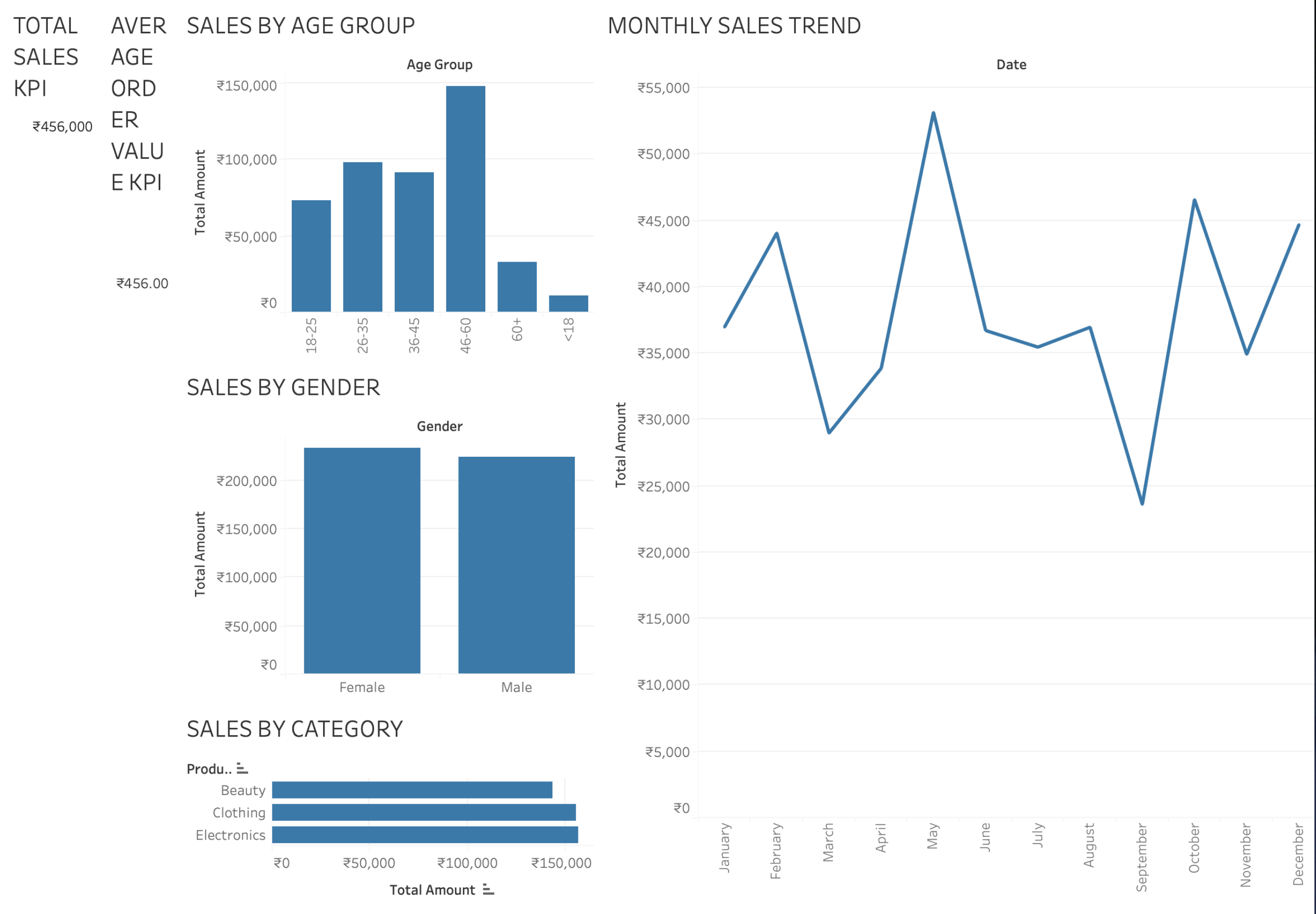
- Sales performance was evaluated across different product categories.
- **Electronics** contributes the highest revenue due to higher-priced products.
- **Clothing** shows high sales volume driven by frequent customer purchases.
- Lower-performing categories highlight opportunities for targeted promotions and improvement.

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Customer Demographics

- Customer purchasing behavior was analyzed based on age and gender.
- Female customers contribute slightly higher sales compared to male customers.
- The **26–45 age group** represents the most valuable customer segment.
- These insights support customer segmentation and targeted marketing strategies.

Dashboard Overview



Key Insights & Inferences

- Overall sales performance shows consistent revenue generation across multiple categories.
- Revenue is driven by both high-value products and high-volume sales segments.
- Key customer groups contribute a significant share of total sales.
- Sales trends indicate the influence of seasonality and purchasing patterns.

Conclusion

- The project provides a comprehensive understanding of retail sales performance.
- Data visualization helps identify important trends and patterns efficiently.
- The analysis supports informed and data-driven business decisions.

Business Value

- Improves decision-making through clear and interactive visual insights.
- Supports inventory optimization and product performance analysis.
- Enables targeted marketing using customer segmentation insights.
- Reduces manual reporting and improves operational efficiency.